

The self.tiles variable :

Is a list of list of tiles. It is the basic representation of the board. Each tile has an index of row and column [row, col]. We can always access the tiles using the self.tiles. But sometimes we need the groups to make checking the tiles values easier.

A_1	B_1	C_1	D_1	E_1	F_1	G_1	H_1	I_1
A_2	B_2	C_2	D_2	E_2	F_2	G_2	H_2	I_2
A_3	B_3	C_3	D_3	E_3	F_3	G_3	H_3	I_3
A_4	B_4	C_4	D_4	E_4	F_4	G_4	H_4	I_4
A_5	B_5	C_5	D_5	E_5	F_5	G_5	H_5	I_5
A_6	B_6	C_6	D_6	E_6	F_6	G_6	H_6	I_6
A_7	B_7	C_7	D_7	E_7	F_7	G_7	H_7	I_7
A_8	B_8	C_8	D_8	E_8	F_8	G_8	H_8	I_8
A_9	B_9	C_9	D_9	E_9	F_9	G_9	H_9	I_9

The self.groups variable :

Is a list of list of tiles. It should have a group (list of tiles) aliasing each row, column block. Tiles should be present in three groups each (once in a row, once in a column and once in a block). Also, each group must be distinct. That is there should be no two groups that have the same tiles.

A_1	A_2	A_3	A_4	A_5	A_6	A_7	A_8	A_9
B_1	B_2	B_3	B_4	B_5	B_6	B_7	B_8	B_9
...	...	...	...	...	...	...	...	...
I_1	I_2	I_3	I_4	I_5	I_6	I_7	I_8	I_9

A_1	B_1	C_1	D_1	E_1	F_1	G_1	H_1	I_1
A_2	B_2	C_2	D_2	E_2	F_2	G_2	H_2	I_2
...	...	...	...	...	...	...	...	...
A_9	B_9	C_9	D_9	E_9	F_9	G_9	H_9	I_9

A_1	B_1	C_1	A_2	B_2	C_2	A_3	B_3	C_3
...	...	...	...	...	...	...	...	...
G_7	H_7	I_7	G_8	H_8	I_8	G_9	H_9	I_9